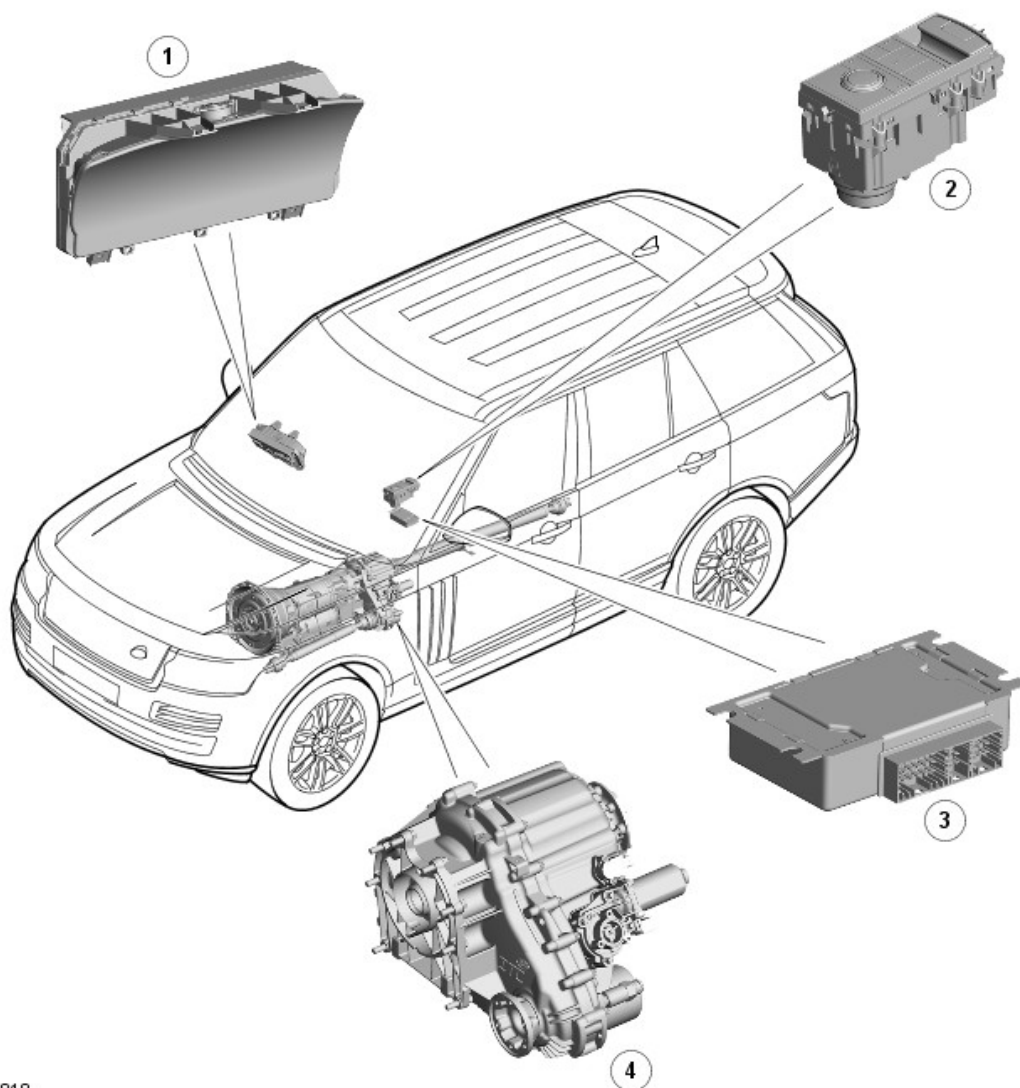


2016.0 RANGE ROVER (LG), 308-07

TRANSFER CASE - VEHICLES WITH: TWIN SPEED TRANSFER CASE

DESCRIPTION AND OPERATION

COMPONENT LOCATION



E148818

ITEM	DESCRIPTION
1	Instrument cluster
2	Terrain Response switchpack
3	Transfer Case Control Module (TCCM)
4	Transfer case

OVERVIEW

The DD295 transfer case is full time, permanent four-wheel-drive unit, with 50/50 torque distribution to the front and rear driveshafts. The transfer case is identical for all engine derivatives and supports the following features:

- Permanent four-wheel-drive with a bevel gear center differential, providing a 50:50 torque split
- Selectable high and low range for optimum on-road and off-road performance
- Two-speed, fully synchronized 'shift-on-the-move' system allows the driver to change the range without having to stop the vehicle
- Electronically controlled multi-plate clutch providing a center differential lock and torque biasing function to give improved traction performance and vehicle dynamic stability.

NOTE:

On vehicles from early 2013, the 'shift on the move' system only applies to range changes from Low to High range. Range changes from High to Low range can only be made when the vehicle is stationary on these vehicles. Transfer Case Control Modules (TCCM) for these vehicles are not compatible with earlier vehicles with full 'shift on the move' transfer case.

A strategy, to electronically control the center differential multi plate clutch assembly, has been developed to provide;

- a pre-loading function, increasing locking torque with increased driving torque
- a slip controller to increase locking torque under off-road conditions and decrease locking torque for optimum comfort, e.g. parking.

The unit is located under the vehicle and is mounted on the cross-member, behind the transmission.

The transfer case receives a torque input from the transmission output shaft, which is passed through the unit to two outputs for the front and rear drive shafts.

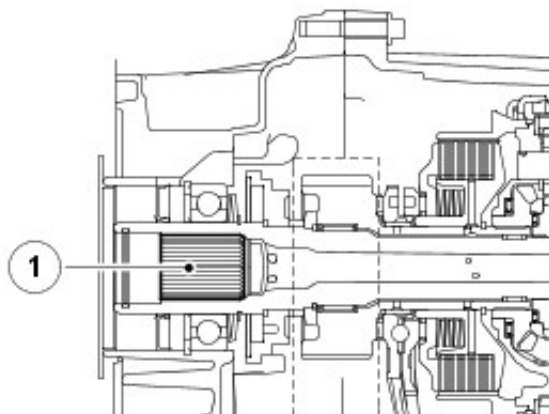
The input torque is equally distributed via a bevel gear type differential. In order to provide an optimal torque distribution to each wheel in all driving conditions, the transfer case is equipped with an electronically controlled locking and torque-biasing device. This device detects wheel slip via various vehicle system inputs to the Transfer Case Control Module (TCCM) and locks the differential accordingly. The locking torque is applied through a multi-plate clutch assembly.

A planetary gear set, located in the differential assembly, allows the driver to select high or low range whilst driving, this is known as 'shift on the move'. When in low range, the planetary gear set provides a ratio of 2.93:1, which gives the vehicle an extremely low crawl speed for off road driving and trailer towing. High range is a direct drive from the transmission output shaft and provides a 1:1 ratio.

Both the center differential locking and biasing and the 'shift on the move' features are actuated via a DC (direct current) transfer case motor, which is controlled by the TCCM, via a PWM (pulse width modulation) signal.

The oil used in the transfer case is specially developed and contains unique additives, which enhance the transfer case operation. No other oil must be used in the transfer case.

A unique type of grease, Weicon anti-seize montagepaste grau TL 7391, must be applied to the transfer case input shaft spline when installing the transfer case.

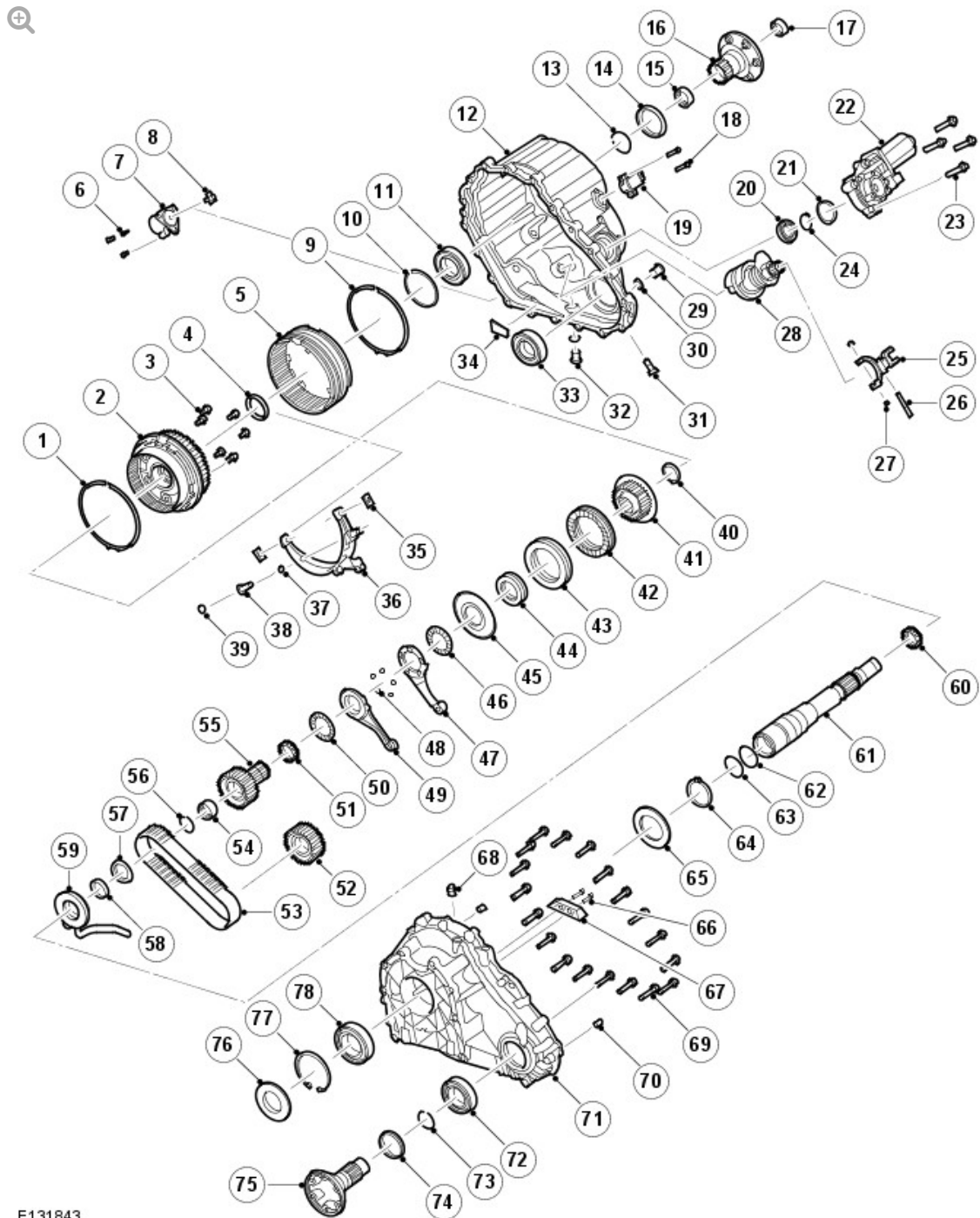


E49212

ITEM	DESCRIPTION
1	Input shaft spline

DESCRIPTION

Transfer Case Exploded View



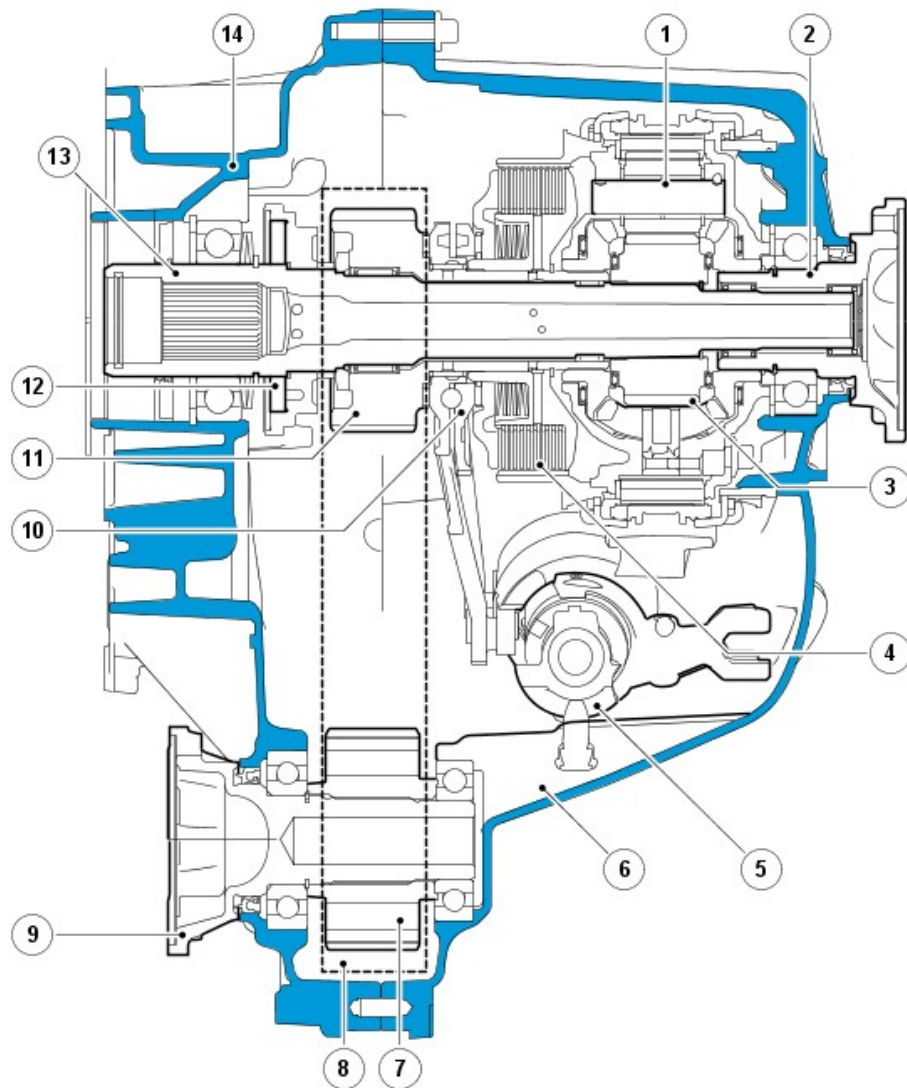
E131843

ITEM	DESCRIPTION
1	Synchronisation spring
2	Differential assembly
3	Bolt, 6 off
4	Spacer ring
5	Shifting sleeve
6	Bolt, 3 off
7	Solenoid
8	Shifting element
9	Synchronisation spring
10	Circlip
11	Ball bearing
12	Rear casing
13	Circlip
14	Seal ring
15	Needle roller bearing
16	Rear output flange
17	Needle roller bearing
18	Bolt, 2 off
19	Selector fork position sensor
20	Bearing
21	Circlip
22	Transfer case motor assembly
23	Bolt, 4 off
24	Circlip
25	Shifting fork
26	Fork pin
27	Sliding block

28	Actuator assembly
29	Fill plug
30	Seal ring
31	Ball retention
32	Drain plug
33	Seal ring
34	Particle collector magnet
35	Sliding blocks, 2 of
36	High/low shifting fork
37	O-ring
38	High/low fork pin
39	Circlip
40	Circlip
41	Clutch hub
42	Clutch friction plate, 10 off
43	Clutch steel plate, 10 off
44	Disc spring, 6 off
45	Clutch piston
46	Axial needle roller bearing
47	Transfer case motor lever assembly
48	Ball, 5 off
49	Transfer case motor lever assembly
50	Axial needle roller bearing
51	Needle roller bearing
52	Front output sprocket
53	Chain
54	Needle roller bearing
55	Sprocket

56	Circlip
57	Thrust washer
58	Spacer ring
59	Oil pump assembly
60	Needle roller bearing
61	Input shaft
62	Retaining ring
63	O-ring
64	Circlip
65	Shim
66	Screw (2 off)
67	Snubber
68	Breather cartridge
69	Bolt, 19 off
70	Dowel pin (2 off)
71	Front casing
72	Bearing
73	Circlip
74	Seal ring
75	Front output flange
76	Seal ring
77	Circlip
78	Bearing

Transfer Case - Sectional View

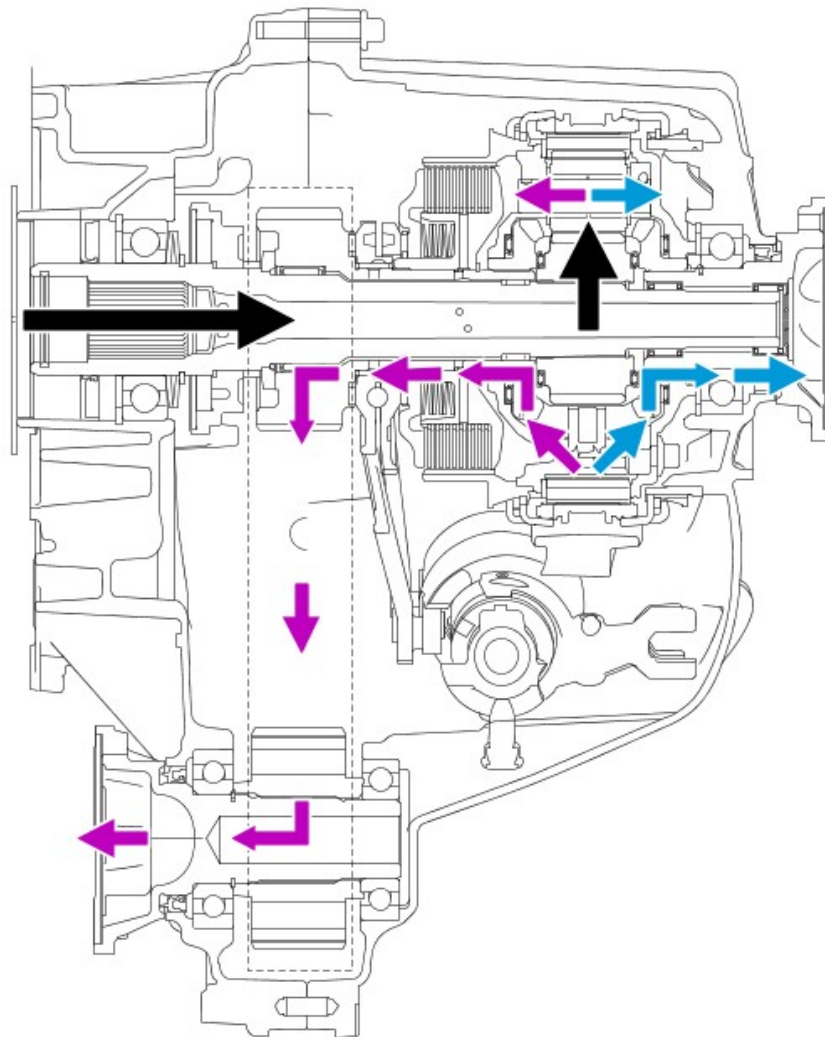


E49191

ITEM	DESCRIPTION
1	Planetary gear set
2	Rear output flange
3	center differential assembly
4	Multi-plate clutch
5	Transfer case motor
6	Rear housing assembly
7	Front output sprocket
8	Chain drive
9	Front output flange
10	Transfer case motor levers

11	Sprocket
12	Oil pump assembly
13	Input shaft
14	Front housing assembly

Transfer Case Power Flow



E49192

Input torque from the transmission is transferred to the input shaft of the transfer case and then onto the planetary sun gear and planetary pinion gears. The planetary pinion gears are held in place by the planet pinion shafts, which are connected to the differential carrier, and drive the differential pinion gears. The torque is then distributed to both the front and rear carriers, which are connected to the outputs of the transfer case.

The rear carrier is connected directly to the rear output flange; the front carrier is connected to the sprocket and therefore to the chain drive, which provides front output flange rotation.

FRONT CASING ASSEMBLY

The front casing assembly provides the location for the input shaft bearing, the front output flange bearing and the oil pump. It is also equipped with threaded holes to mount the chassis mounting bush, 2 lifting eyes and a breather cartridge for the transfer case breather pipe. The breather pipe allows an equalization between atmospheric and internal transfer case pressure.

The front casing has an additional boss cast on the inside. The boss is used to mount a snubber which is secured with 2 screws. The snubber suppresses chain vibrations which can occur under medium acceleration conditions, therefore improving NVH (noise, vibration and harshness) issues. During normal operation the chain does not contact the snubber.

REAR CASING ASSEMBLY

The rear casing assembly provides the location for the rear output flange bearing, the transfer case motor and the oil fill and drain plug. Fins are cast into the rear casing assembly to improve heat dissipation. The unit number is also stamped into the rear housing.

OIL PUMP

An oil pump assembly is located in the front casing to provide lubrication for the bearings and rotary components through cross-drillings in the input shaft. A flat-sectioned coupling on the input shaft drives the rotor of the pump; the stator is fixed to the front housing assembly. A tube is attached to the pump, which leads into a calm suction area at the bottom of the 2 casing assemblies. The collector magnet in the suction area of the pump collects any metallic debris.

CHAIN DRIVE

The chain-drive transfers drive from the center differential to the front output flange. A 3/8" pitch chain connects the sprocket on the transfer case

input shaft with the sprocket on the front output flange. As both sprockets have the same number of teeth, the rotational speed of both sprockets is identical.

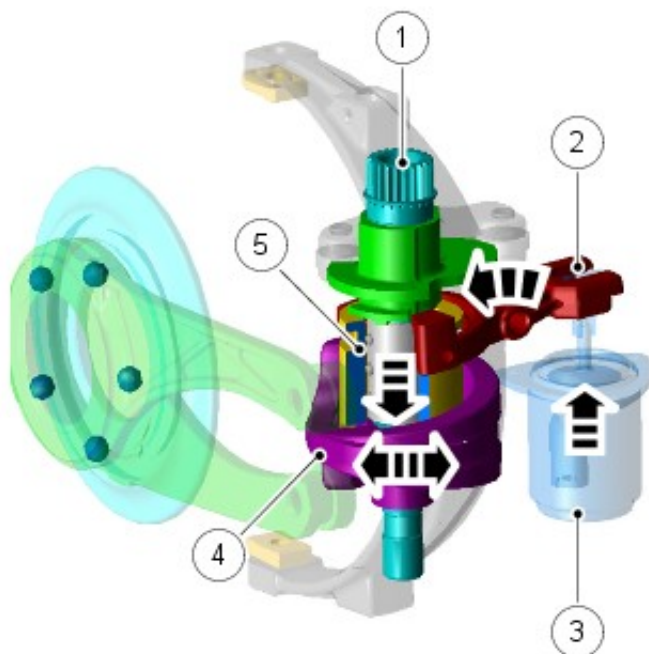
NOTE:

When installing or replacing the chain, the BLUE guide links must be facing the clutch pack, towards the rear output flange.

TRANSFER CASE MOTOR

One motor operates both the high/low range change and the differential locking and torque-biasing device (multi-plate clutch). The motor solenoid switches between the 2 functions, while the motor provides the rotational movement for both operations.

TRANSFER CASE MOTOR POSITION FOR CLUTCH CONTROL MODE

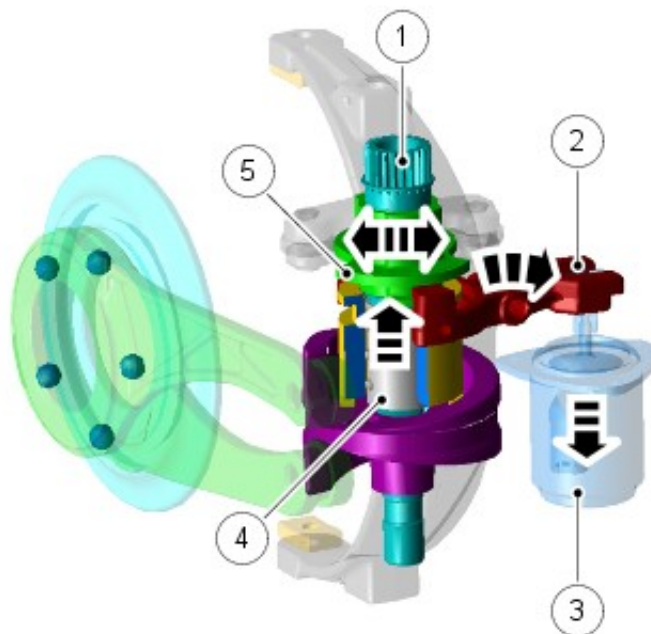


ITEM	DESCRIPTION
1	Motor shaft
2	Solenoid shift fork
3	Solenoid
4	Clutch control disc
5	Shifting sleeve

To actuate the multi-plate clutch, the TCCM energizes the solenoid. The solenoid pin pivots the solenoid shift fork, which engages the shifting sleeve into the dog teeth on the clutch control disc. The rotational movement of the motor shaft is then linked to the clutch control disc via the shifting sleeve.

This is the normal operating mode of the transfer case. In this position, the range change function is disengaged and mechanically locked.

TRANSFER CASE MOTOR POSITION FOR HIGH/LOW RANGE MODE

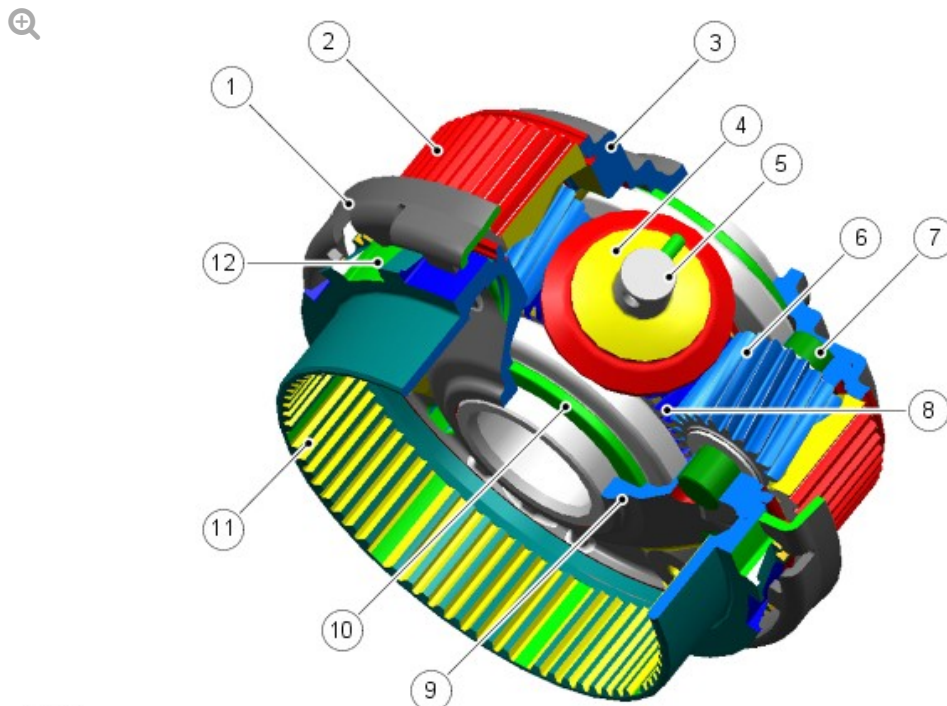


ITEM	DESCRIPTION
1	Motor shaft
2	Solenoid shift fork
3	Solenoid
4	Shifting sleeve
5	Actuation cam

To actuate the high/low range change, the TCCM de-energizes the solenoid. A spring in the solenoid retracts the solenoid pin and rotates the solenoid shift fork. This engages the shifting sleeve to the dog teeth on the high/low actuation cam. The rotational movement of the motor shaft is then linked to the cam.

In this position, the multi-plate clutch is open, the differential cannot be locked and torque cannot be biased. Once the range change is complete the system returns to clutch control mode. In the event of an electrical failure, the motor will default to this position.

CENTER DIFFERENTIAL ASSEMBLY



ITEM	DESCRIPTION
1	Synchronisation cup and spring
2	Planetary ring gear
3	Differential carrier
4	Pinion gears
5	Pinion gear shafts
6	Planetary pinion gears
7	Planetary pinion gear shafts
8	Planetary sun gear
9	Differential cover
10	Differential side gears
11	Multi-plate clutch basket
12	Dogteeth

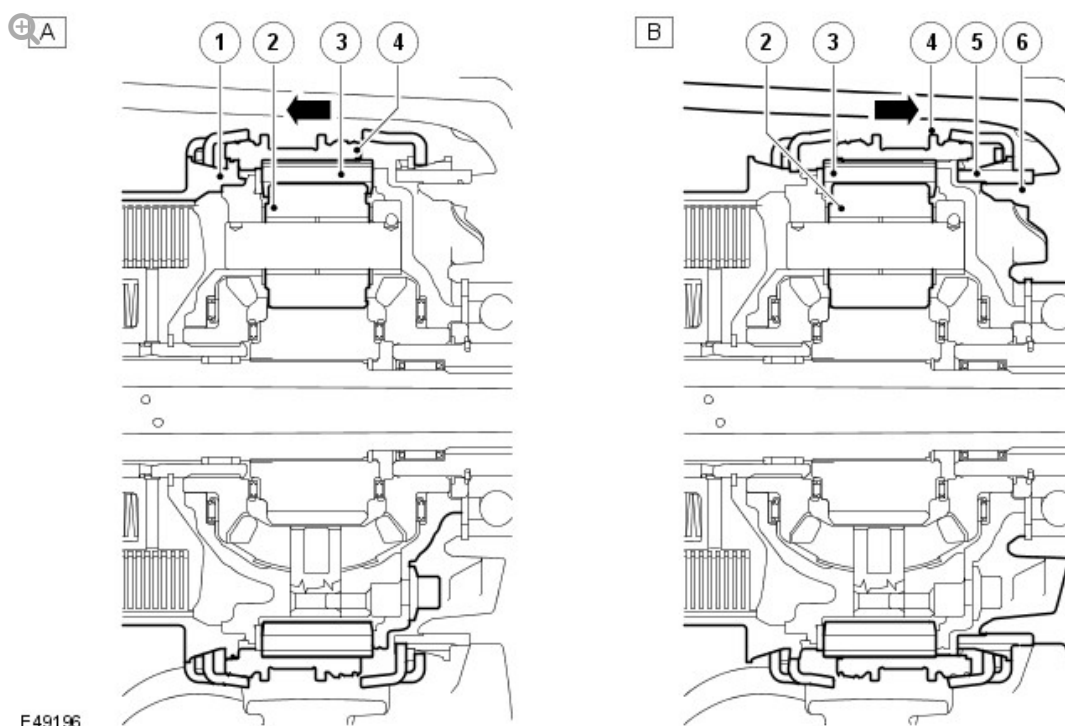
The center differential assembly is the primary feature of the transfer case. Torque is transmitted through the center differential carrier and distributed to the differential gears and the front and rear output flanges. The planetary gear set, for the high/low range change function, is also an integral part of the center differential assembly.

The assembly comprises 3 differential pinion gears and shafts, which are equally spaced within the center differential carrier. The differential shafts have a rigid connection to the differential carrier. Located between the pinion gears are 3 planetary pinion gears and shafts. The planetary sun gear and 2 differential side gears are located in the center line of the carrier.

The planetary ring gear is supported in both directions by the differential casing and the differential cover. The planetary ring gear is connected to a shifting sleeve, which is engaged in either high or low range.

The multi-plate clutch basket, which is welded to the differential casing,

supports the friction plates, the dogteeth for high range engagement and the synchronisation cup and spring for the 'shift-on-the-move' function.

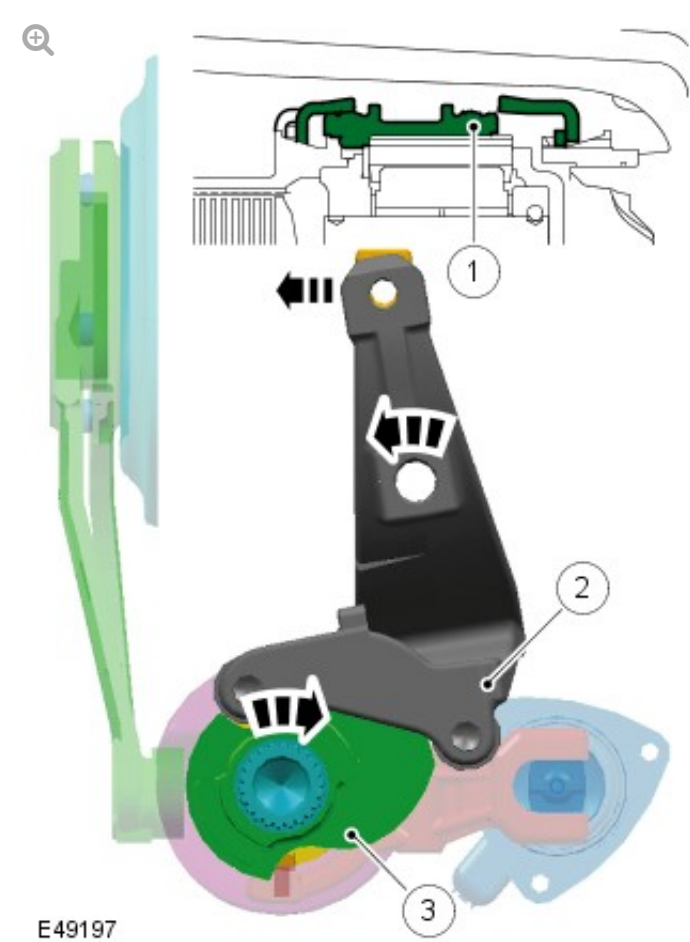


ITEM	DESCRIPTION
A	High range position
B	Low range position
1	Dogteeth
2	Planetary pinion gears
3	Planetary ring gear
4	Shifting sleeve
5	Low range dogteeth
6	Rear carrier assembly

When high range is engaged, the shifting sleeve connects to the differential carrier via dog teeth. The planetary ring gear, via the shifting sleeve, and the planetary pinion gears, via the planetary shafts, which are also attached to the differential carrier. The planetary gear set rotates as one unit and therefore turns the differential side gear with a 1:1 ratio.

In low range, the motor moves the shifting sleeve in the direction of the low range dog teeth. The low range dog teeth, with the synchronisation cup and spring, are fixed to the rear carrier assembly. When the shifting sleeve is engaged with the low range dogteeth, the planetary ring gear, via the shifting sleeve, is stationary and the planetary pinion gears, via the planetary bolts, turn the differential side gears with 2.93: 1 ratio.

HIGH RANGE ACTUATION SEQUENCE

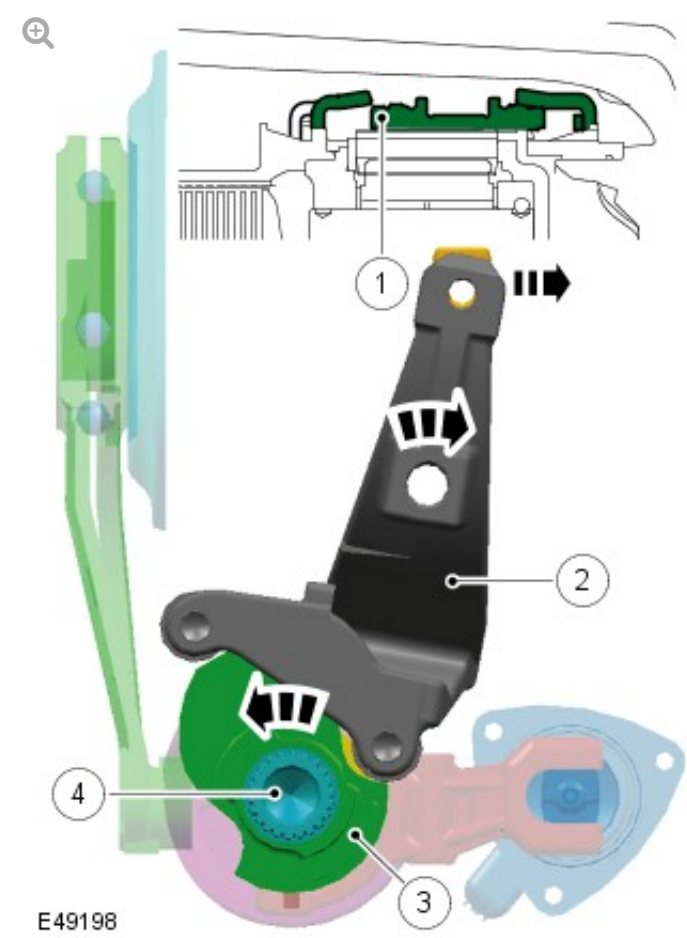


ITEM	DESCRIPTION
1	Shifting sleeve
2	High/low shifting fork
3	Shifting cam

The rotational movement of the motor shaft turns the shifting cam to high range position. The shifting cam then moves the shifting sleeve, via the high/low shifting fork, into the high range position. After the

synchronisation sequence, the planetary ring gear is connected to the high range dog teeth, via the shifting sleeve, on the differential carrier. In this position, the input speed equals the output speed, which equates to a high range ratio of 1:1.

LOW RANGE ACTUATION SEQUENCE

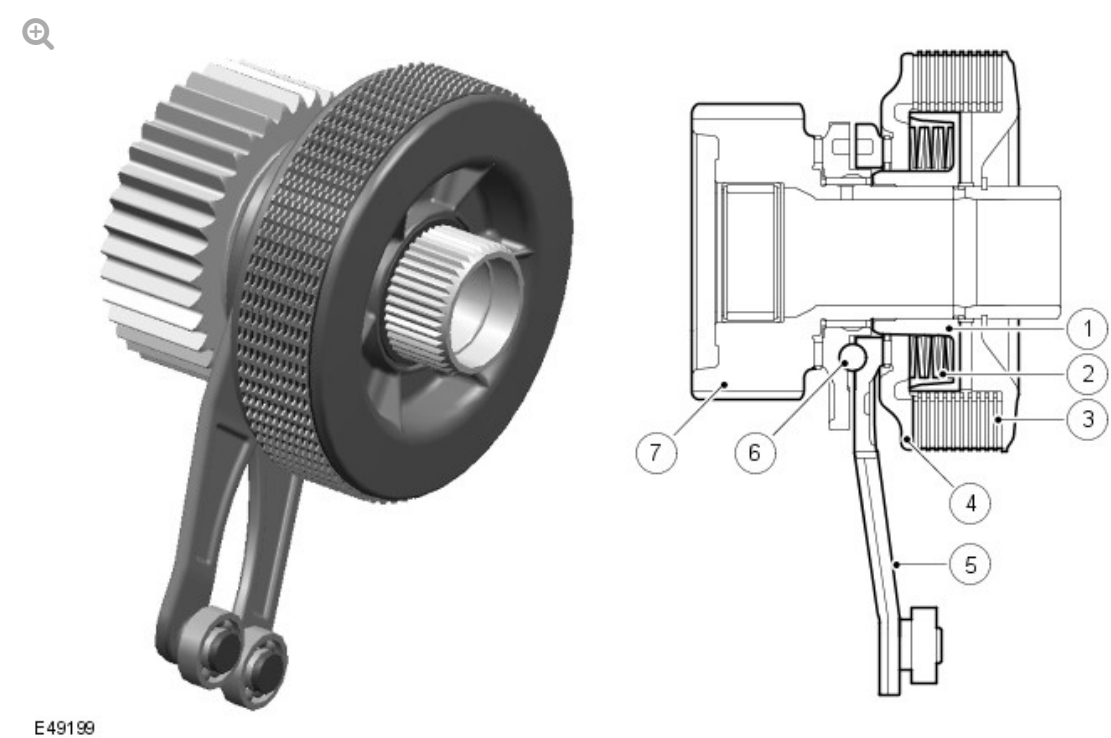


ITEM	DESCRIPTION
1	Shifting sleeve
2	High/low shifting fork
3	Shifting cam
4	Motor shaft

The rotational movement of the motor shaft turns the shifting cam into low range position. The shifting cam then moves the shifting sleeve of the center differential assembly via the high/low shifting fork into low range

position. After the synchronisation sequence, the planetary ring gear is connected to the low range dog teeth, via the shifting sleeve, on the rear carrier assembly. The output speed is then reduced to a ratio of 2.93:1.

MULTIPLATE CLUTCH ASSEMBLY



ITEM	DESCRIPTION
1	Clutch hub
2	Cup springs
3	Clutch plates
4	Clutch piston
5	Motor levers
6	Ball ramp mechanism
7	Sprocket

The multiplate clutch assembly for both center and rear differentials act in a similar way. The aim of the multiplate clutch assembly is to prevent excessive differential slip and therefore maximize the traction performance of the vehicle. This is fundamentally different from the 'braked' traction

control, which can only counteract differential slip when it occurs.

A certain amount of differential slip is required to allow the vehicle to turn corners and to remain stable under control of the ABS (anti-lock brake system). The TCCM monitors the driver's demands through primary vehicle controls and automatically sets the slip torque at the differentials. The system is completely automatic and does not require any special driver input.

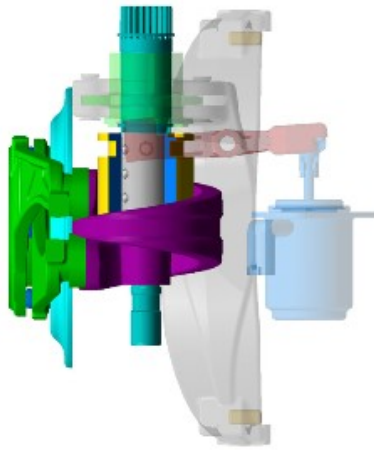
The multiplate clutch assembly actively controls the torque flow through the center differential and optimizes the torque distribution in the driveline. The clutch assembly biases the torque from the transmission to the axle and wheels with the higher grip and prevents the wheels with the lower grip from spinning.

The multiplate clutch assembly comprises the sprocket, which is connected to the front differential side gear, the motor levers with the ball ramp mechanism, the clutch hub as support for the clutch plates, the clutch piston to generate friction between the clutch plates, and a pack of cup springs to return the clutch piston into its original position.

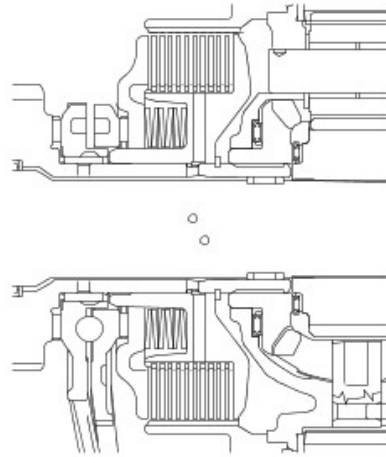
One set of friction plates are connected to the clutch hub; the other set of friction plates are connected to the multi-plate clutch basket, which is welded to the center differential housing.

MULTIPLATE CLUTCH ACTUATION

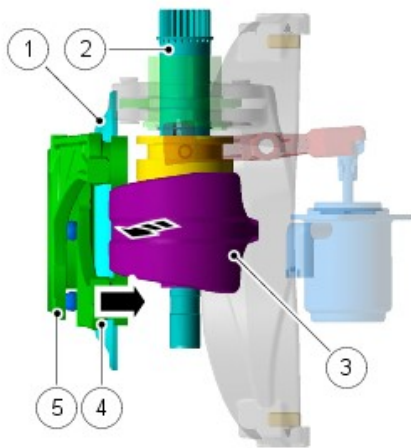
Transfer Case Motor Levers in initial position, multiplate clutch open condition



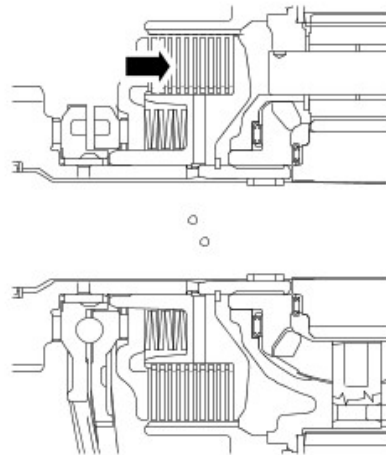
E49200



Transfer Case Motor in end position, multiplate clutch closed condition



E49201

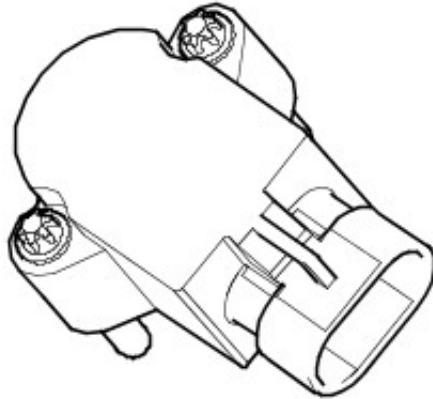


ITEM	DESCRIPTION
1	Clutch piston
2	Motor shaft
3	Clutch control disc
4	Motor levers
5	Ramp mechanism balls

By turning the clutch control disc, via the motor shaft, the motor levers are rotated relative to each other. This relative movement acts on 5 balls in a ramp mechanism between the 2 levers and give a defined axial movement.

The movement forces the clutch piston to induce friction between the plates supported by the clutch hub and the plates supported by the clutch basket on the differential carrier. This frictional force inhibits the differential rotation; the differential carrier and front differential side gear are locked together.

HIGH/LOW RANGE POSITION SENSOR

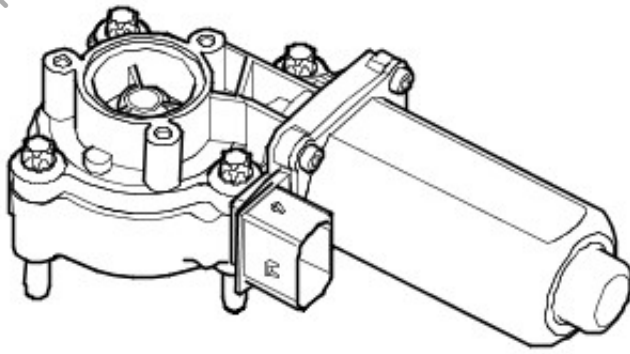


E49209

The high/low range position sensor converts the pivotal movement of the high/low fork into a PWM signal on the input. The PWM signal of the position sensor differs between high range and low range. The TCCM checks this signal and informs the driver, via the instrument cluster and the range change switch LED (light emitting diode), if a range change is in progress or has been completed.

The high/low range position sensor is connected to the TCCM via a 3-pin connector.

TRANSFER CASE MOTOR



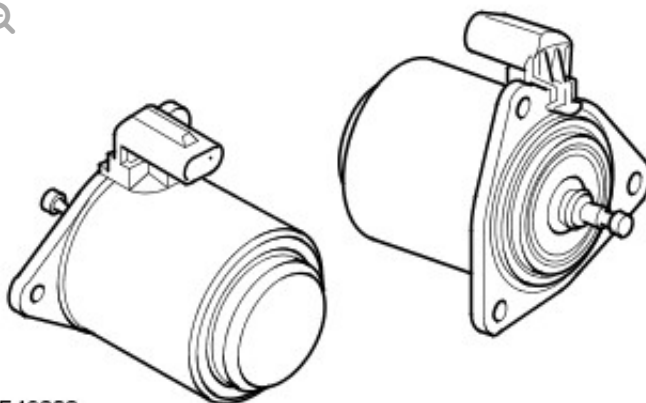
E49207

The transfer case motor provides the necessary movement to perform the high/low range change and the multiplate clutch actuation. The motor is located on the rear casing assembly and secured with four bolts.

The motor is a PWM controlled, DC motor with an integrated worm gear reduction drive. It is connected to the TCCM with an 8-pin connector; the power supply of the motor is maintained through 2 large diameter cables on the motor connector. An internal position sensor checks the rotational movement of the motor.

There is a temperature sensor located within the motor housing. the sensor is connected to the TCCM.

SOLENOID



E49208

The solenoid switches the power flow on the actuation system between high/low range change mode and clutch control mode. When the solenoid is energized, the solenoid pin deploys and activates the clutch control

mode. When the solenoid is de-energized, the internal spring retracts the solenoid pin and activates the high/low range change mode.

NOTE:

In order to replace the solenoid in service, the solenoid must be energized using the Land Rover approved diagnostic system.

The solenoid is connected to the TCCM with a 2-pin connector.

RANGE STATUS INDICATION

The low range status indicator has an LED illuminated mountain symbol on the switch and has the following logic:

- LED on = low range
- LED off = high range
- LED flashing = range change in progress/range undefined/range fault.

There will also be a message displayed in the message center, which can also inform the driver of any faults with the transfer case.

The following table shows the messages that can be displayed in the message center relating to the transfer case:

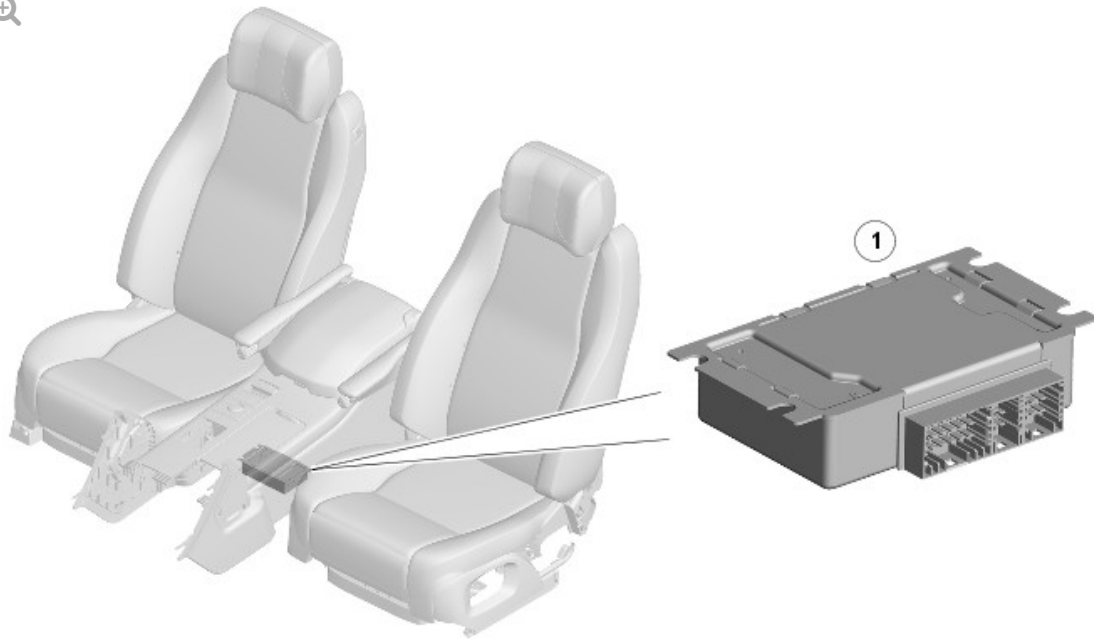
MESSAGE	DESCRIPTION	CHIME
'LOW RANGE SELECTED'	Transfer case has engaged low range after a range change request	None
'HIGH RANGE SELECTED'	Transfer case has engaged high range after a range change request	None
'SPEED TOO HIGH FOR RANGE CHANGE'	Range change request when vehicle speed too high	Single
'SELECT NEUTRAL FOR RANGE CHANGE'	Range change request when lever not in neutral	Single
'APPLY PARK	This alerts the driver that the transmission park lock	Continuous

BRAKE' (PARKING BRAKE US/CAN)	function is inoperative due to transfer case out of high or low range.	
	TCCM has stopped transmitting on the CAN bus during a range change or while in neutral and as a result the transmission park lock function is inoperative.	
'GEARBOX OVERHEATING'	Center differential temperature is approaching the over heated threshold.	None
'GEARBOX FAULT'	Center differential has failed - operating as an open differential.	None
	TCCM has stopped transmitting on the CAN bus and defaults to open center differential.	

The following conditions may generate a message and/or a warning indicator and may be accompanied by a chime:

TRANSFER CASE CONTROL MODULE (TCCM)

The TCCM controls the high/low 'shift-on-the-move' actuation and the multiplate clutch actuation. The TCCM is located below the floor console stowage compartment.



E148820

The TCCM is connected to the high speed CAN (controller area network) chassis systems and controls transfer case operation using high speed CAN messages from other system control modules. The TCCM receives a permanent power supply via a 30A fusible link located in the EJB (engine junction box), and an ignition supply via a 5A fuse and the ignition relay in the CJB (central junction box).

The TCCM memorizes the position of the transfer case motor when the ignition is off (power mode 0).

The TCCM uses the same actuator to control both range change functions and application of center differential locking torque. The TCCM uses position feed back from the actuator to provide smooth range changing capability and graduated application of locking torque appropriate for the current driving conditions. Range change can be carried out while moving providing the transmission is in neutral and the vehicle is below the speed necessary for the requested range change.

The TCCM uses 3 connectors for all inputs and outputs.

The TCCM uses a series of programmed shift maps to control the synchronization speed and ensure that a maximum shift time of

approximately 1 second is achieved.

If the TCCM is replaced, the Land Rover approved diagnostic system must be connected to the vehicle and the TCCM self-calibration procedure must be performed. This procedure must also be performed if the transfer case motor assembly is replaced.

DEFAULT/LIMP-HOME STRATEGY

If a fault occurs with the transfer case, the TCCM or one of the required input signals (for example road speed signal), the TCCM records a DTC (diagnostic trouble code) and will respond appropriately to provide the highest level of system capability under the specific fault conditions. The following fault states are possible:

FAULT STATE	SYSTEM RESPONSE	DRIVER WARNING
No reduction in capability	Diagnostic Trouble Code (DTC) will be recorded but no effect on performance	None
Clutch control not possible. Temporary over temperature condition	The tractive capability of the vehicle, off road, is reduced	Warning indicator and/or 'GEARBOX OVERHEATING' message in message center
Clutch control not possible. Permanent fault	The tractive capability of the vehicle, off road, is reduced	Warning indicator and/or 'GEARBOX FAULT' message in message center
Stuck in Transfer case neutral	The transfer case is stuck between high and low range resulting in no drive to wheels	Flash low range indicator plus 'APPLY PARK BRAKE' (PARKING BRAKE US/CAN) message in message center to apply electric park brake. The message will only display at times when it is deemed safe or necessary for the electric park brake to be applied. It will not display during normal driving for example

If a driveline over temperature condition has occurred, after the driveline has been allowed to cool, clutch control will be re-enabled and the warnings will disappear. There is no need to seek service assistance following an over temperature event.

If two range changes are attempted (high range to low range and low range to high range) within 1 minute and ambient temperature is below 0°C (32°F), the TCCM will inhibit further range changes for 1 minute. This is to protect the transfer case due to the high current required to change range due high viscosity of the transmission oil in cold temperatures. This is correct operation and is not due to a fault, a warning indicator or message will not be displayed.

If clutch control or range change is not possible due to a permanent fault, the driver must seek service assistance at the earliest opportunity.

If the system suffers a fault, which causes the transfer case to fail in neutral, the TCCM is designed to continue attempting to engage the requested range or return to its original range for a fixed number of attempts. If this has not been successful and the low range LED is still flashing, the driver should bring the vehicle to a halt and attempt the range change again while stationary. If this does not work after a number of attempts, turn the ignition off (power mode 0) for 30 seconds, restart the engine and request range change again while stationary. The driver must seek service assistance at the earliest opportunity.

TRANSFER CASE CONTROL MODULE INPUTS/OUTPUTS

The TCCM receives the following inputs:

- Range change selection switch
- High/low position sensor
- Transfer case motor temperature
- Transfer case motor position sensor
- High speed CAN bus messages.

The TCCM sends the following outputs:

- High speed CAN bus messages
- High/low range change LED
- Transfer case motor
- Solenoid.

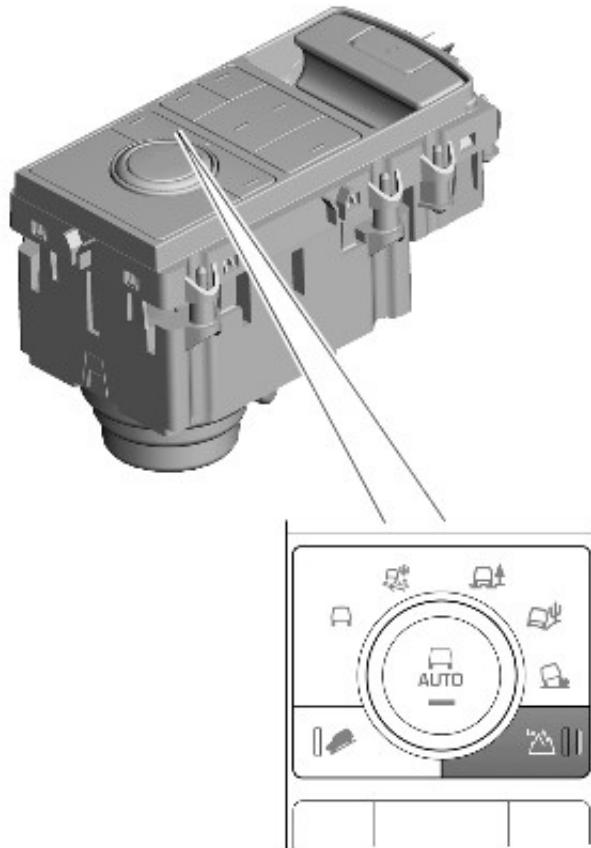
CAN MESSAGES

The TCCM is connected on the high speed CAN chassis systems and controls transfer case operation using high speed CAN messages received via the gateway module from other system control modules. Wheel speed, vehicle acceleration, engine torque and speed, gear information from the transmission, temperature information, car configuration, axle ratios and Terrain Response® mode inputs, are some of the main signals received by the TCCM.

In the event of a high speed CAN bus failure, the following symptoms may be observed:

- Shift from high to low or low to high inoperative
- Instrument cluster low range warning lamp inoperative
- Warning messages or indicator lamps displayed in instrument cluster.

RANGE CHANGE SELECTION SWITCH



E148819

The range change selection switch is located in the Terrain Response (TR) switchpack, in the floor console. The switch is a momentary switch. A single press of the switch will initiate the range change, the transfer case will then change to the other available range. A second press of the switch will revert back to the previously selected range.

When the switch is pressed, it completes a momentary connection to 12V with the TCCM. The TCCM receives this momentary signal and attempts a range change.

The switch is also connected to the high speed CAN bus to the TCCM for control of the switch LED.

HIGH/LOW RANGE CHANGE LED

The LED is connected to the Terrain Response switchpack and the TCCM is responsible for illuminating the range change LED in the switch via the high speed CAN bus.

When the transfer case is in high range, the switch LED is not illuminated.

When changing range, the LED will flash on until the new range status has been achieved.

When changing range the LED will start flashing only when the range change has commenced (for example speed and neutral conditions have been met). If low range is selected, the LED will be illuminated continuously once the change to low range is complete. The LED flash rate is 2 Hz with a 50% duty cycle.

The LED has 2 levels of intensity, high when the vehicle lights are switch off and low when they are switched on.

If LED is flashing at 0.5 Hz, this would indicate a transfer case fault or that the transfer case is in undefined range and may require calibration.

OPERATION

HIGH RANGE OPERATION

In high range, the torque input from the transmission is passed to the transfer case input shaft. The position of the synchronizer sleeve couples the shaft directly to the differential housing. The differential splits the torque between the 2 side gears. One side gear is connected by splines and passes the torque to the rear output flange. The second side gear is connected to the chain drive sprocket and passes the torque, via the chain, to the front output flange.

LOW RANGE OPERATION

In low range, the torque input from the transmission is passed to the transfer case input shaft. The synchronizer sleeve is moved and connects the planetary carrier to the differential housing. The torque from the transmission is now directed through the sun gear of the epicyclic gearset and, via the pinion gears and pinion gear shafts, into the planetary carrier. The annulus gear of the epicyclic gearset is secured inside the casing and

generates the low range ratio of 2.93:1. The torque is then passed, via the synchronizer sleeve, to the differential housing where it is split between the 2 side gears. One side gear is connected by splines and passes the torque to the rear output flange. The second side gear is connected to the chain drive sprocket and passes the torque, via the chain, to the front output flange.

HIGH/LOW RANGE WITH SHIFT-ON-THE-MOVE

NOTE:

On vehicles from early 2013, the 'shift on the move' system only applies to range changes from Low to High range. Range changes from High to Low range can only be made when the vehicle is stationary on these vehicles.

The driver is able to change between high and low range gears while the vehicle is moving, or if the vehicle is stationary.

Pressing the range change switch makes a range change request to the TCCM.

The driver is informed of the range status via a green warning indicator lamp in the instrument cluster and the LED in the range change switch. The warning indicator lamp is not illuminated in high range, but is illuminated in low range and flashes during a range change.

The vehicle will remain in the selected range unless the driver requests a change, i.e. it will not automatically revert to high range following an ignition off/on sequence.

RANGE CHANGE PROCEDURE

NOTE:

On vehicles from early 2013, the 'shift on the move' system only applies to range changes from Low to High range. Range changes from High to Low range can only be made when the vehicle is stationary on these vehicles.

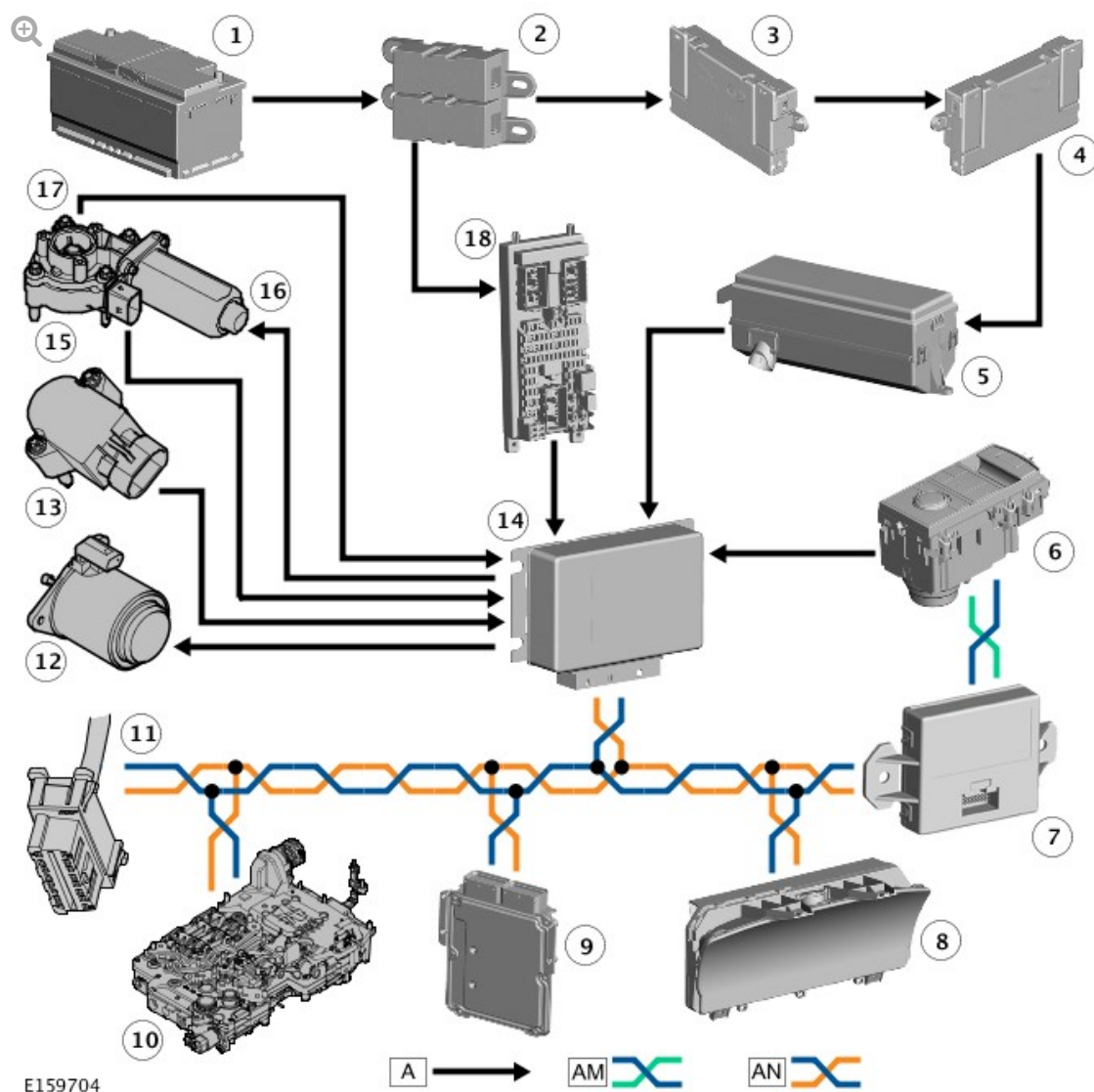
Neutral must be selected on the transmission before requesting a range change and then select the appropriate gear following completion of the range change. During the range change the transmission will be locked in neutral.

If neutral is not selected when a range change is requested, then the request is denied and the driver will be advised to select neutral via the instrument cluster message center.

The range change process can take several seconds to complete following a request being accepted. The actual time is dependent on a number of factors; including the transfer case temperature, component wear and if the first attempt was unsuccessful.

There is a limit set on the maximum speed at which a range change can be achieved. The maximum speed for a high to low range change is 25 mph (40 km/h). On vehicles with low to high 'shift on the move', the maximum speed for a low to high range change is 37 mph (60 km/h). If the vehicle speed is over the limit when a range change is selected, the request is denied and the instrument cluster message center will display a 'SPEED TOO HIGH FOR RANGE CHANGE' message.

CONTROL DIAGRAM



A = HARDWIRED; AM = HIGH SPEED LIN (LOCAL INTERCONNECT NETWORK) CHASSIS SYSTEMS; AN = HIGH SPEED CAN POWERTRAIN SYSTEMS.

ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box 2 (BJB2)
3	Battery Junction Box (BJB)
4	Auxiliary Junction Box (AJB)
5	Engine Junction Box (EJB)
6	Terrain Response® Switchpack (TRS) (incorporating range switch)
7	Gateway Module (GWM)

8	Instrument cluster
9	Engine Control Module (ECM)
10	Transmission Control Module (TCM)
11	Diagnostic socket
12	Solenoid
13	High/Low range position sensor
14	Transfer Case Control Module (TCCM)
15	Temperature sensor
16	Transfer case motor
17	Transfer case motor Hall sensors (speed and direction)
18	Central Junction Box (CJB)